WORKED SOLUTION

STEP III 2021, Question 8

Pure • Recurrence Relations and Limits

STEP III 2021, Question 8

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 2.8, *Squaring and Subtracting Two*: https://beyondstep.keithmcnulty.org/ch02/squaring-minus-two.html

The key idea

Everything rests on factorising the recurrence. Subtracting 2 gives $x_{n+1} - 2 = (x_n - 2)(x_n + 2)$, so once the terms are above 2 their distance from 2 is multiplied by more than 4 at every step. That gives part (i), and part (ii) follows. For part (iii), subtracting 4 from x_{n+1}^2 factorises just as neatly, and that is exactly what makes the product $x_1 x_2 \cdots x_n$ telescope.

Solution

Part (i)

We use induction on n. For $n = 1$ the claim reads $x_1 \geq 2 + 4^0(a-2) = a$, which is true with equality (note $4^0 = 1$).

Suppose that $x_n \geq 2 + 4^{n-1}(a-2)$ for some $n \geq 1$. Since $a > 2$, the right-hand side is greater than 2, so $x_n > 2$. Now

$$x_{n+1} - 2 = x_n^2 - 4 = (x_n - 2)(x_n + 2).$$

Here $x_n - 2 > 0$ and $x_n + 2 > 4$, so $(x_n - 2)(x_n + 2) \geq 4(x_n - 2)$. Using the inductive hypothesis $x_n - 2 \geq 4^{n-1}(a-2)$,

$$x_{n+1} - 2 \geq 4(x_n - 2) \geq 4 \cdot 4^{n-1}(a-2) = 4^n(a-2),$$

which is the claim for $n+1$. By induction, $x_n \geq 2 + 4^{n-1}(a-2)$ for all $n \geq 1$.

We never squared an inequality. Squaring $x_n \geq (\text{bound})$ is only valid when both sides are non-negative, and the factorised form avoids that worry entirely.

Part (ii)

If $|a| > 2$. Then $x_2 = a^2 - 2 > 4 - 2 = 2$. The sequence $x_2, x_3, \ldots$ obeys the same recurrence and starts above 2, so part (i), applied with x_2 in place of a (shifting the index by one), gives

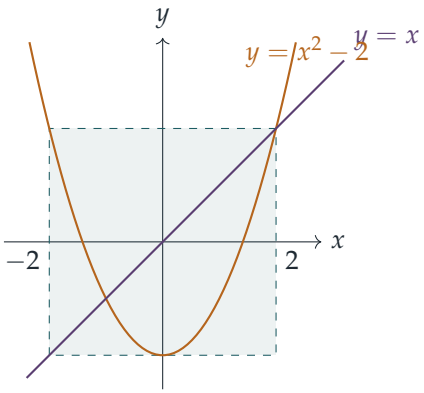
$$x_n \geq 2 + 4^{n-2}(x_2 - 2) \qquad (n \geq 2).$$

Since $x_2 - 2 > 0$ and $4^{n-2} \to \infty$, we get $x_n \to \infty$.

If $|a| \leq 2$. We show by induction that $-2 \leq x_n \leq 2$ for every n. This holds for $n = 1$. If $-2 \leq x_n \leq 2$ then $0 \leq x_n^2 \leq 4$, so $-2 \leq x_n^2 - 2 \leq 2$, which means $-2 \leq x_{n+1} \leq 2$. So the sequence is bounded and cannot tend to infinity.

? **Ask yourself** In the case $|a| \leq 2$, is it enough to show that $x_n \leq 2$ for all n?

No. An upper bound alone is not enough, because a term below -2 would square to something larger than 4, and the following term would be above 2. That is why the induction has to carry both bounds together. The graph shows why it works: the parabola $y = x^2 - 2$ maps the interval $[-2, 2]$ onto itself.



Putting the two cases together, $x_n \to \infty$ if and only if $|a| > 2$.

Part (iii)

The factorisation that drives everything here is

$$x_{n+1}^2 - 4 = (x_n^2 - 2)^2 - 4 = x_n^4 - 4x_n^2 = x_n^2\,(x_n^2 - 4).$$

Since $a > 2$, part (i) gives $x_n > 2$ for all n, so $x_n > 0$ and $x_n^2 - 4 > 0$. Taking positive square roots,

$$\sqrt{x_{n+1}^2 - 4} = x_n \sqrt{x_n^2 - 4}.$$

Applying this repeatedly (formally, by induction on n),

$$\sqrt{x_{n+1}^2 - 4} = x_n \sqrt{x_n^2 - 4} = x_n x_{n-1} \sqrt{x_{n-1}^2 - 4} = \cdots = x_1 x_2 \cdots x_n \sqrt{a^2 - 4}.$$

Therefore, for every $n \geq 1$,

$$\frac{\sqrt{x_{n+1}^2 - 4}}{x_{n+1}} = \frac{\sqrt{a^2-4}\, x_1 x_2 \cdots x_n}{x_{n+1}},$$

and comparing with the definition of y_n shows that the required formula holds for all n exactly when $A = \sqrt{a^2-4}$.

Finding a. The value $A = \sqrt{a^2-4}$ is forced. At $n = 1$ the required formula reads $Aa/x_2 = a\sqrt{a^2-4}/x_2$, and since $a > 0$ and $x_2 > 0$ this gives $A = \sqrt{a^2-4}$. Conversely, $A = \sqrt{a^2-4}$ with $a > 2$ means $A^2 = a^2 - 4$, so

$$a = \sqrt{A^2+4}.$$

This value does satisfy $a > 2$, because $A > 0$ gives $A^2 + 4 > 4$. With this a we have $\sqrt{a^2-4} = A$, and the displayed identity above becomes $y_n = \sqrt{x_{n+1}^2 - 4}\,/\,x_{n+1}$ for all $n \geq 1$, as required.

Convergence. With this value of a, since $x_{n+1} > 0$,

$$y_n = \frac{\sqrt{x_{n+1}^2 - 4}}{x_{n+1}} = \sqrt{1 - \frac{4}{x_{n+1}^2}}.$$

As $a > 2$, part (ii) gives $x_{n+1} \to \infty$, so $4/x_{n+1}^2 \to 0$ and $y_n \to 1$. The second sequence converges, and its limit is 1.

Checks

- Take $A = 1$, so $a = \sqrt{5} \approx 2.236$. Then $x_2 = 3$, $x_3 = 7$ and $y_1 = \sqrt{5}/3 \approx 0.745$. From the formula, $\sqrt{9-4}/3 = \sqrt{5}/3$, which agrees. Next, $y_2 = \sqrt{5} \cdot 3/7 \approx 0.958$ and $\sqrt{49-4}/7 = 3\sqrt{5}/7$, which also agrees. The terms $y_3 \approx 0.99909$ and $y_4 \approx 0.9999996$ are racing towards 1.
- The substitution $a = t + 1/t$ with $t > 1$ gives $x_n = t^{2^{n-1}} + t^{-2^{n-1}}$. This shows how fast the growth really is (doubly exponential, far faster than the 4^{n-1} of part (i)), and it is consistent with every result above.
- The bound in part (i) and the boundedness in part (ii) were also tested by computer on many random starting values.

Common slips

- Squaring the inequality from the inductive hypothesis without first checking that both sides are non-negative.
- In part (ii), bounding only $x_n \leq 2$ when $|a| \leq 2$. You also need $x_n \geq -2$, or the next term could jump above 2.
- Working backwards to find a but never confirming that this value works for all n and satisfies $a > 2$.

Beyond STEP

The map $x \mapsto x^2 - 2$ is the most famous member of the quadratic family $x \mapsto x^2 + c$. On $[-2, 2]$, putting $x = 2\cos\theta$ turns it into angle-doubling, $\theta \mapsto 2\theta$, a standard first example of chaotic dynamics. Outside that interval every orbit escapes to infinity. For $a > 2$ the same idea works with $x = 2\cosh u$, and then the product in part (iii) is a hyperbolic version of the classical $\prod \cos(\theta/2^k)$ telescoping. For the analysis behind the limits and inductions here, read Abbott, *Understanding Analysis*.

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